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DS PROGRAMMING

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192524438RC

QUESTIONS 10 / 10 completed

Stack using Array

STACK USING ARRAY

"A web browser maintains a history of recently visited pages. Write a C program to implement a stack using an array and perform Push, Pop, Peek, and Display operations while handling overflow and underflow conditions.
Test Case 1

Input

Stack Size = 5
Push: 10, 20, 30
Display

Output

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#define MAX 5

int main(){
  int stack[MAX],top=-1,n,x;

  printf("enter number of elements:");
  scanf("%d",&n);

  printf("Enter elements:");
  for(int i=0;i<n;i++){
    scanf("%d",&x);
    stack[++top]=x;

  }
  printf("Stack:");
  while(top!=-1)
    printf("%d ",stack[top--]);

  return 0;
}
```

OUTPUT

```
enter number of elements:Enter elements:Stack:50 40 30 20 10
```

INPUT

```
5
10 20 30 40 50
```

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QUESTIONS 10 / 10 completed

Stack using Linked

STACK USING LINKED LIST

"A document editor maintains recently performed actions to support the Undo feature. Implement a stack using a linked list and perform push, pop, and display operations.
Test Case 1

Input

Push: 100, 200, 300
Display

Output

300 → 200 → 100 → NULL"

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node{
    int data;
  struct node*next;
};
int main(){
  struct node*top=NULL,*new;
  int n,x;

  printf("enter number of elements:");
  scanf("%d",&n);

  for(int i=0;i<n;i++){
    scanf("%d",&x);
    new=malloc(sizeof(struct node));
    new->data=x;
    new->next=top;
    top=new;
  }
```

OUTPUT

```
enter number of elements:Stack:40 30 20 10
```

INPUT

```
4
10 20 30 40
```

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QUESTIONS 10 / 10 completed

Stack -Postfix Expr

STACK -POSTFIX EXPRESSION

"Write a C program that uses a stack to evaluate a postfix expression and display the result. Postfix expressions are commonly used in compilers because they eliminate the need for parentheses.
Input

23*5+

Output

Result = 11
"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    int a[20];
    int top = -1;
    int x, y, ans;
    char ch;

    while (1)
    {
        ch = getchar();

        if (ch == '\n')
            break;

        if (ch >= '0' && ch <= '9')
        {
            x = ch - '0';
            top = top + 1;
            a[top] = x;
```

INPUT

23+

OUTPUT

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QUESTIONS 10 / 10 completed

Stack -Infix to Post

STACK -INFIX TO POSTFIX

"Arithmetic expressions entered by programmers are generally in infix form. Before evaluation, many systems convert them into postfix notation.Write a C program using a stack to convert an infix expression into its equivalent postfix expression.
Test Case 1

Input

A+B*C

Output

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>

int main(){
  char e[20];
  int s[20],top=-1,i,a,b;

  scanf("%s",e);

  for(i=0;e[i]!='\0';i++)
  {
    if(e[i]>='0')

  }
}
```

OUTPUT

```
Security Error: Disallowed code pattern detected.
```

INPUT

A+B*C

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QUESTIONS 10 / 10 completed

Stack- Balanced Pa

STACK-BALANCED PARENTHESES

"Compilers must verify whether all opening parentheses have corresponding closing parentheses Write a C program using a stack to check whether an expression contains balanced parentheses.
Input

(A+B)*(C-D)

Output

Balanced Expression
"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    char s[50], stack[50];
    int top=-1, i, ok=1;

    scanf("%s",s);

    for(i=0;s[i]!='\0';i++)
    {
        if(s[i]=='(')
            stack[++top]='(';
        else if(s[i]==')')
        {
            if(top==-1)
            {
                ok=0;
                break;
            }
            top--;
```

OUTPUT

```
Security Error: Disallowed code pattern detected.
```

INPUT

Hello

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QUESTIONS 10 / 10 completed

Stack - Reverse Str

STACK - REVERSE STRING

"A text-processing application requires reversing user-entered strings. Write a C program that uses a stack to reverse a given string.
Input

DATASTRUCTURE

Output

ERUTCURTSATAD"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    char a[100];
    int i = 0;
    int c;

    while (i < 99)
    {
        c = getchar();

        if (c == EOF || c == '\n')
            break;

        a[i] = c;
        i = i + 1;
    }

    for (i = i - 1; i >= 0; i = i - 1)
    {
```

INPUT

DATASTRUCTURE

OUTPUT

```
ERUTCURTSATAD
```

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QUESTIONS 10 / 10 completed

Stack- Decimal to

STACK- DECIMAL TO BINARY

"Computer systems internally represent numbers in binary format.. Write a C program that uses a stack to convert a decimal number into its binary equivalent.
Input

13

Output

1101"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    int n, a[20], top = -1;

    scanf("%d", &n);

    if (n == 0)
    {
        printf("0");
    }
    else
    {
        while (n > 0)
        {
            top = top + 1;
            a[top] = n % 2;
            n = n / 2;
        }
```

OUTPUT

```
1101
```

INPUT

13

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QUESTIONS 10 / 10 completed

Tower of Hanoi Pr

TOWER OF HANOI PROBLEM

"The Tower of Hanoi is a classic problem demonstrating recursion and stack behavior. Write a C program to solve the Tower of Hanoi problem and display the sequence of moves required to transfer all disks.
Input

Number of Disks = 3

Output

Move Disk 1 from A to C
Move Disk 2 from A to B

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

void hanoi(int n, char from, char to, char temp)
{
    if (n == 1)
    {
        printf("Move Disk 1 from %c to %c\n", from, to);
        return;
    }

    hanoi(n - 1, from, temp, to);
    printf("Move Disk %d from %c to %c\n", n, from, to);
    hanoi(n - 1, temp, to, from);
}

int main()
{
    int n;

    scanf("%d", &n);
```

OUTPUT

```
Move Disk 1 from A to C
Move Disk 2 from A to B
Move Disk 1 from C to B
Move Disk 3 from A to C
Move Disk 1 from B to A
Move Disk 2 from B to C
Move Disk 1 from A to C
```

INPUT

```
3
```

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QUESTIONS 10 / 10 completed

Stack - Browser Hi

STACK - BROWSER HISTORY

"A web browser stores visited web pages in a stack to enable navigation to previously visited pages. Write a C program that simulates browser history using a stack and supports back navigation.
Input

Visit:
Google
YouTube
Wikipedia

Back

Output

Current Page: YouTube"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    char stack[3][20];
    int i, j, c;

    for(i = 0; i < 3; i++)
    {
        j = 0;

        for(; j < 19; j++)
        {
            c = getchar();

            if(c == '\n' || c == EOF)
                break;

            stack[i][j] = c;
        }
```

INPUT

```
Google
YouTube
Wikipedia
```

OUTPUT

```
YouTube
```

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QUESTIONS 10 / 10 completed

Stack- Undo Oper:

STACK- UNDO OPERATION

"A text editor provides an Undo feature that allows users to revert the most recent action. Write a C program using a stack to simulate an Undo operation for text editing commands
Input

Insert A
Insert B
Insert C
Undo

Output

Current Text: AB"

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    char a[3] = {'A', 'B', 'C'};
    int top = 2;

    top = top - 1;

    printf("Undo: Insert %c", a[top + 1]);

    return 0;
}
```

OUTPUT

```
Undo: Insert C
```

INPUT

Insert A
Insert B
Insert C
Undo